

Sunny Raj

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Machine Learning Engineer

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[Github](#)

Data Scientist (6+ Years) specializing in end-to-end AI products across **Healthcare, Insurance, and Retail**. Expert in **Generative AI (RAG, Agentic Workflows)**, MLOps, and scalable data pipelines. Proven track record of deploying production-grade models on **AWS, Azure, and Databricks** to drive measurable business impact.

SKILLS

Generative AI & Agents: RAG Architectures, Agentic AI Workflows, Prompt Engineering, LLM Integration.
Agent Orchestration: LangGraph, Pydantic-AI, Google Agent Development Kit (ADK).
Machine Learning & NLP: Deep Learning, Supervised/Unsupervised Learning, NLP, Neural Networks, Transformers.
Core Data Stack: Python, PySpark, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, SciPy.
MLOps & Cloud: AWS, GCP, Databricks, Snowflake, MLflow, Docker, Kubernetes, CI/CD, Model Monitoring, Flask.
Databases & Search: SQL, NoSQL, **Vector Databases** (Qdrant, ChromaDB), **Azure AI Search**.
UI & Visualization: Streamlit, Chainlit, Plotly, Matplotlib, Seaborn.

WORK EXPERIENCE

Tiger Analytics	MLE - Data Science	Aug2022 - Present
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GenAI & Agentic AI Solutions

- Engineered and deployed a no-code GenAI tool that empowers non-technical users to create and execute custom multi-agent workflows via configuration files, ensuring a modular and scalable architecture.
- Developed reusable tools with Python and integrated them into GenAI applications using the LangChain framework.
- Built and deployed LLM-based applications on Azure, leveraging services like Redis Cache and CosmosDB for robust data management and storage.
- Built an Enterprise RAG Chatbot using Azure AI Search, reducing information retrieval time by 70% and support tickets by 50%.

Global Supply Chain & Consumer Goods

- Designed and deployed scalable data pipelines on AWS EMR using PySpark to process terabytes of data.
- Implemented Apache Airflow to orchestrate dynamic creation of EMR clusters, significantly improving job efficiency.
- Managed and processed large-scale datasets (average size 10+ GB), ensuring data integrity and reliability for business intelligence and reporting.

Healthcare Analytics & MLOps

- Architected and implemented end-to-end data pipelines on Databricks, integrating Unity Catalog and Delta Live Tables for efficient data governance and processing.
- Utilized Kafka for real-time data ingestion and employed MLflow for comprehensive model management, versioning, and deployment of models like XGBoost and CatBoost on AWS.
- Developed ML solutions that enhanced predictive analytics and enabled data-driven decision-making for a major healthcare client.

Finance & Data Engineering

- Automated document validation for student loans using OCR and NLP, building a robust web application deployed on AWS and OpenShift Container Platform (OCP) using Lambda, Step Functions, and EKS.
- Conducted anomaly detection on large datasets and optimized Snowflake database queries, demonstrating a strong focus on data quality and security.

Appyhigh Technology

Machine Learning Engineer

Mar 2020 – Aug 2022

- Engineered and implemented a real-time recommendation system for a news application, handling over 2 million daily API calls and optimizing for accuracy and efficiency.
- Built and deployed a conversational AI chat-bot using NLP techniques with NLTK and spaCy for enhanced user interaction.
- Designed a scalable web scraping infrastructure to collect and process data from millions of web pages daily, utilizing shell scripting and Python to build hundreds of scrapers.
- Developed backend systems for data-driven dashboards, managing MongoDB databases to ensure optimal performance and data integrity.

EDUCATION

Electrical and Electronics Engineering	(2014 - 2018)	(CSVТУ, Bhilai)	- 73.28 %
Higher secondary, PCM	(2011 - 2013)	(C.B.S.E, New-Delhi)	- 75%

PROJECTS

RagForge: Custom MCP RAG & Project Management Agent [Github](#)

- Developed an autonomous, local ReAct RAG agent and project management assistant using LangGraph, MCP servers (Qdrant, OpenProject), and Ollama. Engineered a robust, end-to-end ETL pipeline with Temporal.ai orchestration, integrated with MLflow, Arize Phoenix, and a FastAPI gateway.

Visual Fashion Detector: Classify Fashion Items [Github](#)

- Built a modular, SOLID-compliant Computer Vision framework integrating state-of-the-art detector and segmenter models (YOLOv8, SAM, CLIPSeg) with domain-specific zero-shot classification.
- Developed custom high-performance orchestrators for real-time bounding box generation, transparent PNG cutout extraction, dynamic interactive HTML notebook visualizations, and model benchmarking.

Semantic & Data Projects [Github](#) | [Github](#)

- Employed Sentence Transformers and BERT/roBERTa models for precise keyword extraction based on contextual understanding.
- Built a dynamic real-time parking simulation with a Python-Dash web application deployed on AWS and backed by MongoDB.
- Developed a scalable web scraper to extract and structure data from Google News and Search.

Licenses & certifications

[Databricks Certified Machine Learning Professional](#)

[WS Certified Machine Learning Specialty](#)

[Statistics for Data Science and Business Analysis](#)

[Build a Crypto Bot, 100% Functional - Algorithmic Trading](#)